

**Eighth Semester B. Tech. (Electronics and Communication Engineering) Examination**

**REAL TIME OPERATING SYSTEMS AND KERNELS**

Time : 3 Hours]

[Max. Marks : 60

**Instructions to Candidates :—**

- (1) All the questions are compulsory and carry marks as mentioned.
- (2) Draw neat and clean diagram, wherever necessary.
- (3) Use of Non – programmable calculator is permitted.
- (4) Assume suitable data wherever necessary.

1. (a) List five services provided by an operating system, and explain how each creates convenience for users. In which cases would it be impossible for user-level programs to provide these services ? Explain your answer. 5(CO1)
- (b) What are the models to interface between user-level threads and kernel-level threads ? Under what circumstances is one type better than the other ? Justify your answer. 5(CO1)
2. (a) The certain kernel has a policy that a process cannot hold a spinlock while attempting to acquire a semaphore. Explain why this policy is in place. 5(CO2)
- (b) Consider the following snapshot of a system :

|       | <u>Allocation</u> |   |   | <u>Need</u> |   |   | <u>Available</u> |   |   |
|-------|-------------------|---|---|-------------|---|---|------------------|---|---|
|       | A                 | B | C | A           | B | C | A                | B | C |
| $P_0$ | 0                 | 1 | 0 | 7           | 4 | 3 | 2                | 3 | 0 |
| $P_1$ | 3                 | 0 | 2 | 0           | 2 | 0 |                  |   |   |
| $P_2$ | 3                 | 0 | 2 | 6           | 0 | 0 |                  |   |   |
| $P_3$ | 2                 | 1 | 1 | 0           | 1 | 1 |                  |   |   |
| $P_4$ | 0                 | 0 | 2 | 4           | 3 | 1 |                  |   |   |

Answer the following questions using the banker's algorithm :

- (a) What is the content of the matrix Need ?

- (b) Is the system in a safe state ?
- (c) If a request from process P0 arrives for (1, 0, 0), can the request be granted immediately ? 5(CO2)
3. (a) Differentiate the event registers and signals with examples. 5(CO3)
- (b) Explain why interrupts shall not be disabled frequently and how it affects the system performance. 5(CO3)
4. (a) Explain the difference between internal and external fragmentation. 5(CO4)
- (b) Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB and 125KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB and 375 KB (in order) ? Rank the algorithms in terms of how efficiently they use memory. 5(CO4)
5. (a) Evaluate the performance of the three techniques for allocating disk blocks (contiguous, linked and indexed) for both sequential and random file access. 5(CO4)
- (b) What problems could occur if a system allowed a file system to be mounted simultaneously at more than one location ? 5(CO4)
6. Enlist various structures of RTOS Kernels and their applications. 10(CO5)

